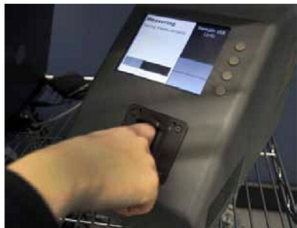


Technology News

Future cars won't let you drink and drive

QinetiQ has developed an alcohol-detection prototype that can instantly detect whether a driver is on booze, without the help of a breathalyser.



QinetiQ alcohol-detection prototype for drivers

The device from QinetiQ has a sensor which non-intrusively analyses the driver's breath and measures blood-alcohol content through the skin with sensors that are placed strategically on the steering wheel and door locks. If the driver has crossed the legal limit of alcohol allowed while driving, the car's engine would not start.

Unlike current alcohol ignition interlock systems, this device doesn't require the driver to blow into a breath-testing device before the car can operate.

Electronic eyeglasses

Bifocals or no-line progressive lenses have a drawback: The lenses that magnify fine print also blur objects more than an arm's length away when the wearer looks down, distorting the view when on a staircase or when swinging at a golf ball, etc.

The electronic spectacles from PixelOptics handle this problem with

an unusual insert in the bottom part of the lenses—liquid crystals, cousins to the familiar ones in television displays. The crystals change how the lenses refract or bend light, just as varying levels of thickness do in traditional glasses.

Called emPower, the glasses come equipped with tiny batteries, microchips and assorted electronics to turn reading power on and off as per your need. To turn on the reading power in the glasses, users need to touch the side of the frame. Batteries in the frame send along a current that changes the orientation of molecules in the crystals. Touch the side of the frame again, and the reading power

disappears.

Portable sensor to warn about asthma attack

By analysing a patient's breath, a new portable sensor from Siemens AG can

tell hours in advance if he will get an asthma attack.

The sensor can detect in the patient's breath minute increases of nitric oxide—particles that serve as a sign of an impending asthma attack. The gas sensor registers if the asthma sufferer's air passages are about to become inflamed, enabling the patient to take anti-inflammatory medication in time to prevent an attack.

The sensor can measure amounts as small as one part per billion (ppb), making it as sensitive as larger devices that are far more expensive and not portable. One ppb corresponds approximately to the dilution of a cube of sugar in a 50 metre-long swimming pool.

Phone with electrocardiogram

With EPI Life mobile phone, you will be able to take an ECG reading of your heart rhythm at home, at work or even in the subway without having to deal with messy gels and electrodes.

The phone, developed in Singapore, has an integrated multi-lead ECG and Health Suite function. It takes your pulse when you press your fingers on a



EPI Life mobile phone that lets you take ECG reading